Logistics Optimization for Delivery Routes – DHL

Project Overview:

DHL, one of the world's leading logistics and courier service providers, operates an extensive global air and ground transportation network, connecting more than 220 countries and territories through major hubs in Leipzig, Cincinnati, Hong Kong, Dubai, and Singapore.

The network includes international sortation centers, regional hubs, and last-mile delivery partners that handle millions of shipments daily across continents.

As global e-commerce volumes continue to surge — especially during peak seasons and holidays — flight delays, customs bottlenecks, and last-mile congestion significantly impact both on-time delivery performance and operational costs.

Currently, DHL's logistics operations face challenges in:

- Identifying the root causes of transit and delivery delays (e.g., customs clearance, weather disruptions, route congestion).
- Optimizing international and regional routes for faster, more cost-effective deliveries.
- Improving shipment efficiency, warehouse throughput, and delivery agent performance using data-driven insights.

The logistics data, stored in relational databases, can be analyzed using SQL to extract meaningful performance metrics and delay patterns.
These insights can help DHL improve hub utilization, route planning, and service-level reliability across its international network.

Project Objective:

Build a SQL-driven logistics analytics system to analyze shipment performance, optimize routes, and enhance delivery efficiency across DHL's global network.
The project aims to:

- Identify delay patterns and operational inefficiencies.
- Optimize hub and route combinations for improved transit times.
- Analyze agent- and warehouse-level performance.

Recommend actionable improvements that strengthen on-time delivery and cost efficiency.
(If you're unfamiliar with any concepts or domain-specific terms, feel free to take the help of Google or AI tools like ChatGPT or Gemini to understand them better and apply them effectively in your project.)

Dataset Description

The dataset will include the following key tables:
(Please download the hyperlinked datasets)

1. Orders Table

The Orders dataset contains order-level delivery details including order date, route, warehouse, and payment type.

Order_ID, Customer_ID, Order_Date, Route_ID, Warehouse_ID, Order_Amount, Delivery_Type, Payment_Mode

2. Routes Table

The Routes dataset includes route-level transportation details, covering the source and destination cities, countries, total distance, and average transit time (in hours).

Route_ID, Source_City, Source_Country, Destination_City, Destination_Country, Distance_KM, Avg_Transit_Time_Hours

3. Warehouses Table

The Warehouses dataset provides location-level information about DHL major hubs and sortation centers.

Warehouse_ID, City, Country, Capacity_per_day, Manager_Name

4. Delivery Agents Table

The Delivery Agents dataset contains agent-level performance data, including agent ID, name, assigned zone and country, experience, and customer rating.

Agent_ID, Agent_Name, Zone, Zone_Country, Experience_Years, Avg_Rating

5. Shipment Tracking Table

The Shipments dataset includes shipment-level tracking information with timestamps for pickup and delivery, delay duration, and feedback ratings.
It captures operational data that helps analyze transit time, on-time delivery percentage, and service-level adherence across routes and agents.

Shipment_ID, Order_ID, Agent_ID, Route_ID, Warehouse_ID, Pickup_Date, Delivery_Date, Delivery_Status, Delay_Hours, Delivery_Feedback

Tasks to be Performed
(Please refrain from using AI to perform the tasks mentioned below, as it will only provide generic solutions.)

Task 1: Data Cleaning & Preparation (10 Marks)

- Identify and delete duplicate Order_ID or Shipment_ID records.
- Replace null or missing Delay_Hours values in the Shipments Table with the average delay for that Route_ID.
- Convert all date columns (Order_Date, Pickup_Date, Delivery_Date) into YYYY-MM-DD HH:MM:SS format using SQL date functions.
- Ensure that no Delivery_Date occurs before Pickup_Date (flag such records).
- Validate referential integrity between Orders, Routes, Warehouses, and Shipments.

Task 2: Delivery Delay Analysis (10 Marks)

- Calculate delivery delay (in hours) for each shipment using Delivery_Date – Pickup_Date.
- Find the Top 10 delayed routes based on average delay hours.
- Use SQL window functions to rank shipments by delay within each Warehouse_ID.
- Identify the average delay per Delivery_Type (Express / Standard) to compare service-level efficiency.

Task 3: Route Optimization Insights (20 Marks)

For each route, calculate:

- Average transit time (in hours) across all shipments.
- Average delay (in hours) per route.
- Distance-to-time efficiency ratio = Distance_KM / Avg_Transit_Time_Hours.
- Identify 3 routes with the worst efficiency ratio (lowest distance-to-time).
- Find routes with >20% of shipments delayed beyond expected transit time.
- Recommend potential routes or hub pairs for optimization.

Task 4: Warehouse Performance (10 Marks)

- Find the top 3 warehouses with the highest average delay in shipments dispatched.
- Calculate total shipments vs delayed shipments for each warehouse.
- Use CTEs to identify warehouses where average delay exceeds the global average delay.
- Rank all warehouses based on on-time delivery percentage.

Task 5: Delivery Agent Performance (10 Marks)

- Rank delivery agents (per route) by on-time delivery percentage.
- Find agents whose on-time % is below 85%.
- Compare the average rating and experience (in years) of the top 5 vs bottom 5 agents using subqueries.
- Suggest training or workload balancing strategies for low-performing agents based on insights.

Task 6: Shipment Tracking Analytics (10 Marks)

- For each shipment, display the latest status (Delivered, In Transit, or Returned) along with the latest Delivery_Date.
- Identify routes where the majority of shipments are still “In Transit” or “Returned”.
- Find the most frequent delay reasons (if available in delay-related columns or flags).
- Identify orders with exceptionally high delay (>120 hours) to investigate potential bottlenecks.

Task 7: Advanced KPI Reporting (10 Marks)

Create SQL queries to calculate and summarize the following KPIs:

- Average Delivery Delay per Source_Country.
- On-Time Delivery % = (Total On-Time Deliveries / Total Deliveries) * 100.
- Average Delay (in hours) per Route_ID.
- Warehouse Utilization % = (Shipments_Handled / Capacity_per_day) * 100.

Use aggregation and CASE statements to compute these metrics and create summarized KPI tables.

Task 8: PPT Presentation (10 Marks)

PPT Submission

- Present your analysis and findings by copying all SQL queries and result tables from Tasks 1 to 7 into a PowerPoint presentation.
- Ensure that tables are formatted clearly and queries are concise.
- Use charts, graphs, and route-based visuals to make your findings more intuitive.
- Include a final slide summarizing key insights and business recommendations for DHL logistics optimization.

Task 9: Video Submission (10 Marks)

Record a video presentation (5–9 minutes**)** explaining:

- Project understanding and approach.
- SQL queries used for analysis.
- Key insights and operational recommendations for DHL.
  (The summary should be in your own words and must not be generated using AI. Please don't write a script and read it aloud. Also, screenshare and show the key findings. Marks will be deducted for failing to do so.)

Note:

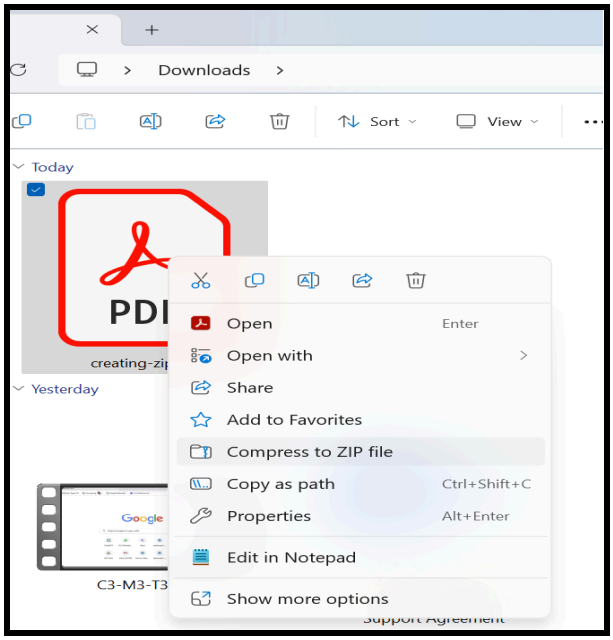
- Submit your PPT and SQL script after zipping those in a folder.
- Plagiarism will result in a penalty, including possible project disqualification.
- The project will be evaluated based on the quality of analysis and visualizations, depth of insights, feasibility of recommendations, clarity of explanations, and adherence to instructions and deliverables.
- If the student sets their own criteria, they need to clearly mention and explain it. Marks will be given according to the specified criteria if they are acceptable.
- Remember to keep your face clearly visible in the video.

Submission Guidelines:

- Save the PPT, SQL Script in a folder and then convert it into a zipped (.zip) folder. **(Please note, the drivelink for the video created should also be added in the Ppt itself.)**
- Upload the zipped folder on your respective dashboard.
- Failure to comply with submission guidelines will result in no grading/0 marks.

How to ZIP a PDF file:

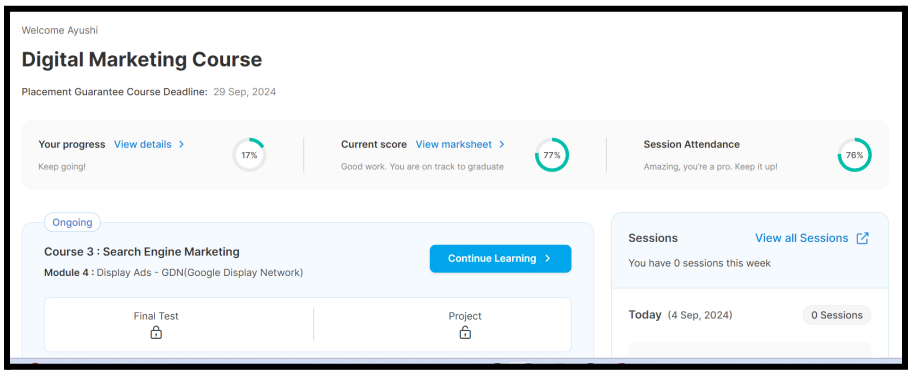
- Put all of the documents you want to compress (or just one) into a new folder.
- Right click on that folder.



- Select the “Compress to ZIP file” option and then click “Compressed (Zipped) folder.”
- A new .ZIP file will be created that contains your document(s).

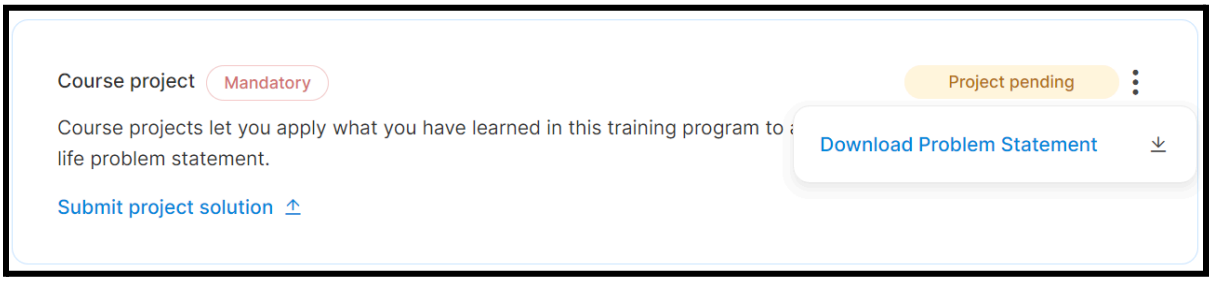
In order to submit the projects please follow the following steps:

1. Click on “ Your progress View details” after logging into your dashboard.



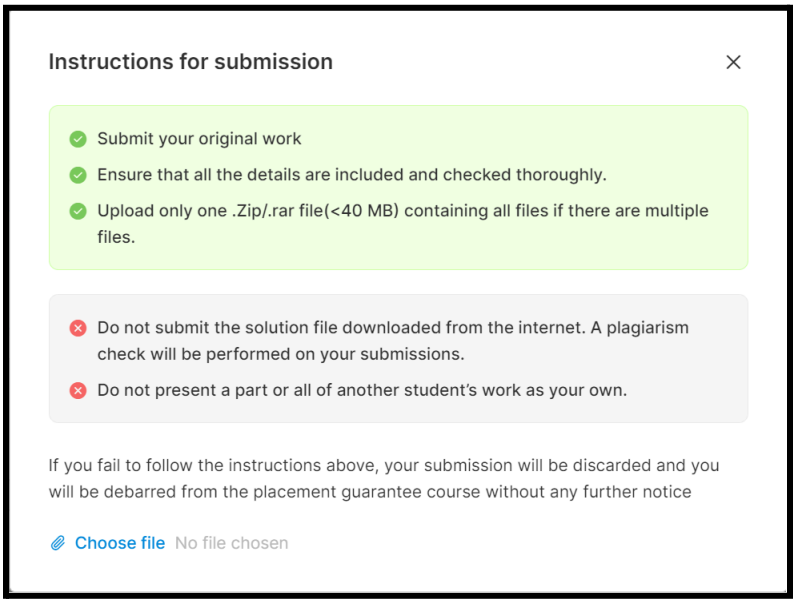
2. Next, click on the tab for the specific child course for which you want to download the problem statement. Then, scroll down to find the "**Course Project**" section.

3. Now, click on the three dots on the right-hand side of the "Course Project" tab to select **"Download Problem Statement."**



4. Please follow the guidelines (screenshot is shared below) provided in the project to ensure correct submissions. Then, click on "**Upload Project Solution**" to submit

your work.


Instructions for submission
Submit your original work
Ensure that all the details are included and checked thoroughly.
Upload only one .Zip/.rar file(<40 MB) containing all files if there are multiple files.
Do not submit the solution file downloaded from the internet. A plagiarism check will be performed on your submissions.
Do not present a part or all of another student's work as your own.
If you fail to follow the instructions above, your submission will be discarded and you will be debarred from the placement guarantee course without any further notice
Choose file
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